

8.3 Classwork: Rational Functions as Transformations

A graphing calculator is allowed; show your reasoning and use the GDC to check sketches. The **parent function** throughout is the reciprocal function $f(x) = \frac{1}{x}$ (asymptotes $x = 0$ and $y = 0$); every function below is built from it by *translations*, *reflections*, and *stretches*.

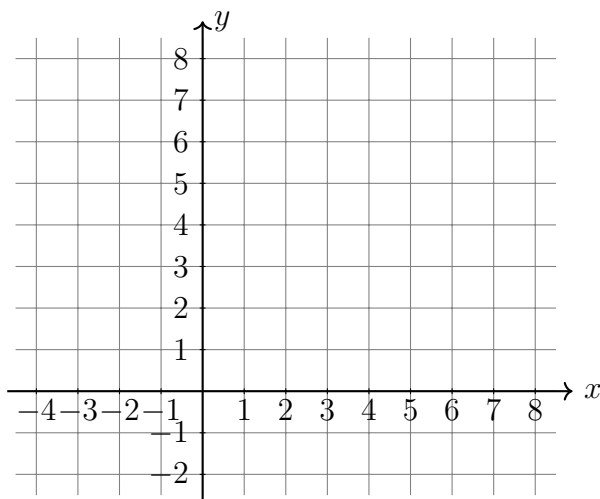
1. **Translations.** Starting from the parent function $f(x) = \frac{1}{x}$:

(a) Describe fully the single transformation that maps f onto $g(x) = \frac{1}{x+4}$.

(b) Describe fully the single transformation that maps f onto $h(x) = \frac{1}{x} + 3$.

(c) The function $p(x) = \frac{1}{x-2} + 5$ is obtained from f by *two* translations. Describe them, and write down the equations of the vertical and horizontal asymptotes of p .

(d) Find the y -intercept of p . Then sketch p on the axes below, showing both asymptotes as dashed lines. Use your GDC to check the shape.



2. **Reflections and stretches.** Again start from $f(x) = \frac{1}{x}$.

(a) Write down the equation of the image of f after a *reflection in the x -axis*.

(b) Write down the equation of the image of f after a *vertical stretch with scale factor 2*.

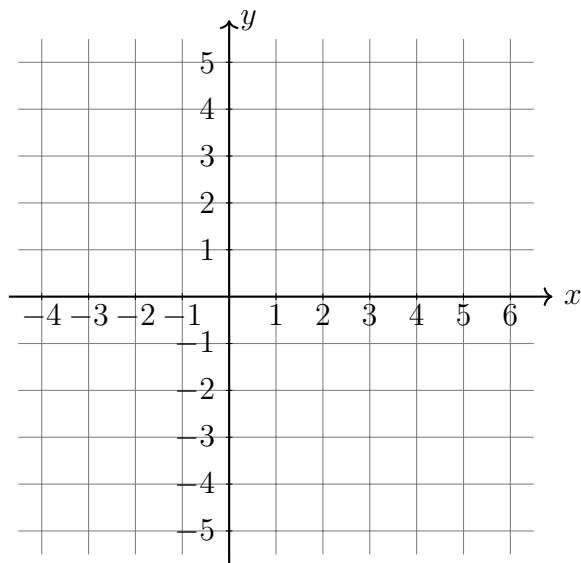
(c) Consider $g(x) = \frac{-2}{x-1}$.

(i) Describe a sequence of transformations that maps f onto g .

(ii) State the equations of the two asymptotes of g .

(iii) Find the coordinates of any axis intercepts of g .

(iv) Sketch g on the axes below (dashed asymptotes); check with your GDC.



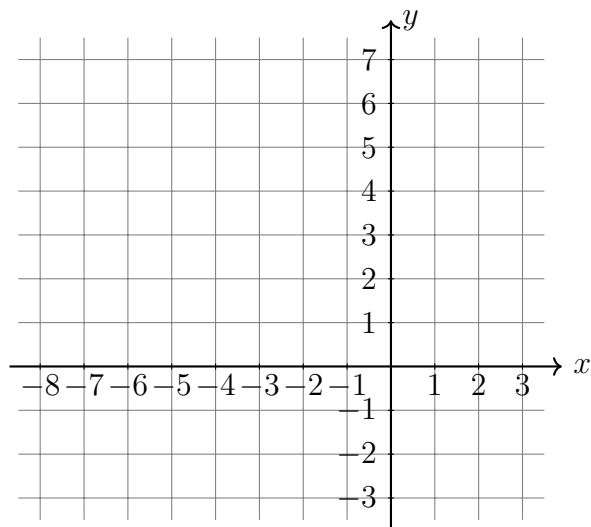
3. **From $\frac{ax+b}{cx+d}$ to a transformation.** Consider $f(x) = \frac{2x+7}{x+3}$, $x \neq -3$.

(a) Use division (or otherwise) to show that $f(x) = 2 + \frac{1}{x+3}$.

(b) Hence describe f as a sequence of transformations of the parent function $y = \frac{1}{x}$.

(c) Write down the equations of the asymptotes, and find the x - and y -intercepts of f .

(d) Sketch f on the axes below. Label the asymptotes (dashed) and the intercepts.



4. **Building a function from transformations.** The graph of a function g is obtained from the parent function $y = \frac{1}{x}$ by the following sequence:

a vertical stretch with scale factor 3, then a translation 2 units right and 1 unit down.

(a) Write down an expression for $g(x)$.

(b) State the equations of the two asymptotes of g .

(c) Find the coordinates of the x -intercept and the y -intercept of g .

(d) Use your GDC to verify your graph, and write down the value of $g(8)$.

5. **Quick identification.** For each function, describe fully the transformation(s) of the parent function $y = \frac{1}{x}$ that produce it, and state the equations of its two asymptotes.

(a) $y = \frac{1}{x - 5}$

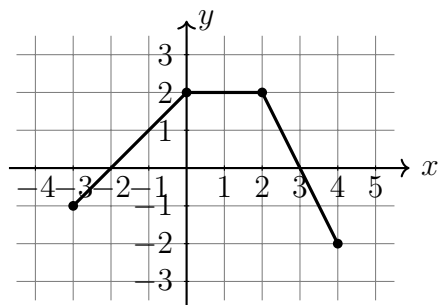
(b) $y = \frac{1}{x} - 4$

(c) $y = \frac{4}{x}$

(d) $y = -\frac{1}{x}$

(e) $y = \frac{1}{x + 2} + 1$

6. **Transforming a given graph.** Transformations work on *any* function, not only the reciprocal. The diagram shows the graph of a function $y = f(x)$, made of three straight segments through the marked points.



- (a) Write down the values of $f(0)$ and $f(2)$.
- (b) The graph of g is obtained from f by the transformation $g(x) = f(x) - 3$. Describe this transformation in words, and write down the coordinates of the image of the point $(0, 2)$.
- (c) On the axes below, sketch the graph of $y = f(x - 1)$ (a translation of f). Mark the images of the four labelled points.

